Grade 11 – Term 1: One Dataset, Many Representations

One worked example: download times (C6 and C7). A school technician records the download times of 20 learning files, in seconds. Only the histogram survives; individual times are unavailable. Convert this one sample into tables and a representative list, then estimate its mean, sample variance and standard deviation. A supplied theoretical model for the process has

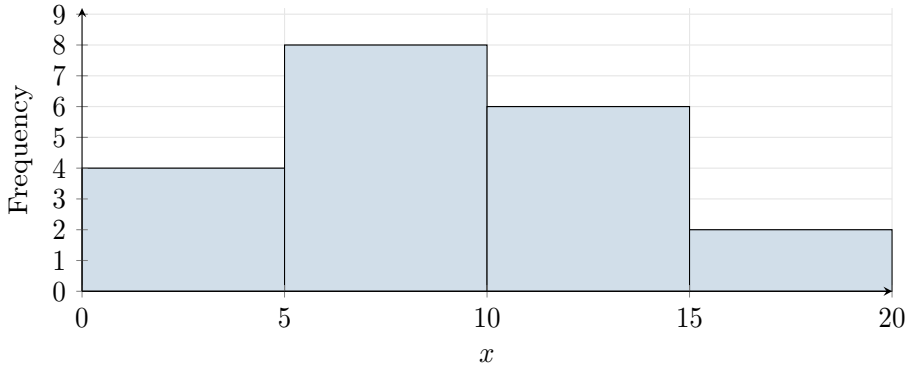
$$\mu = 10 \text{ s}, \qquad \sigma^2 = 25 \text{ s}^2, \qquad P(5 \le X < 15) = 0.80.$$

These are model benchmarks, not results calculated from the sample. The grouped data and probability benchmark follow the download-time context in the reference exam; the mean and variance benchmarks are supplied for this extended example.

Histogram $\longrightarrow$ Interval Table $\longrightarrow$ Midpoint Frequency Table $\longrightarrow$ List of Observations $\longrightarrow$ Statistical Calculations

Color key: intervals; midpoints; frequencies and counts; formulas; substitutions; green boxes for final results. Labels retain these distinctions when printed in grayscale.

1. Read the histogram



Horizontal axis: x = download time (seconds). Vertical axis: frequency (number of files).

Each bar spans five seconds. Read its height from the frequency axis: 4, 8, 6, 2, from left to right. Classes include their left endpoint and exclude their right endpoint: 5 seconds belongs to $[5, 10)$, not $[0, 5)$.

2. Convert to an interval frequency table

Copy each pair of horizontal boundaries and its bar height. No approximation is needed for this transfer.

Class i	Interval (s)	f_i	Reading from the histogram
1	$[0, 5)$	4	Boundaries 0 and 5; height 4.
2	$[5, 10)$	8	Boundaries 5 and 10; height 8.
3	$[10, 15)$	6	Boundaries 10 and 15; height 6.
4	$[15, 20)$	2	Boundaries 15 and 20; height 2.

$$n = \sum_{i=1}^{r} f_i, \qquad r = 4, \quad n = 4 + 8 + 6 + 2 = 20.$$

There are $r = 4$ classes and $n = 20$ observations. Frequencies count files; intervals describe possible times. Neither is a statistical average.

3. Construct the midpoint frequency table

For lower boundary L_i and upper boundary U_i, use the formula

$$m_i = \frac{L_i + U_i}{2}.$$

The midpoint lies halfway between the boundaries. Keep the frequency unchanged.

i	Interval (s)	Midpoint calculation (s)	f_i
1	$[0, 5)$	$(0 + 5)/2 = 2.5$	4
2	$[5, 10)$	$(5 + 10)/2 = 7.5$	8
3	$[10, 15)$	$(10 + 15)/2 = 12.5$	6
4	$[15, 20)$	$(15 + 20)/2 = 17.5$	2

The resulting discrete frequency table is

m_i (s)	2.5	7.5	12.5	17.5
f_i	4	8	6	2

For example, the four original times in $[0, 5)$ could be anywhere within that interval. We represent each by 2.5 seconds. This does not establish that any actual measurement was 2.5, or that the times were uniformly spread in the class.

Approximation begins here. Midpoint replacement preserves all 20 counts but discards variation within intervals. Its statistics are exact for the representative data and estimates for the original observations.

4. Expand into an explicit list of observations

Let y_j denote representative observation j, for $j = 1, \ldots, n$. Write each midpoint m_i exactly f_i times:

$$(y_1, \ldots, y_{20}) = (\underbrace{2.5,\ 2.5,\ 2.5,\ 2.5}_{\text{4 observations}},$$
$$\underbrace{7.5,\ 7.5,\ 7.5,\ 7.5,\ 7.5,\ 7.5,\ 7.5,\ 7.5}_{\text{8 observations}},$$
$$\underbrace{12.5,\ 12.5,\ 12.5,\ 12.5,\ 12.5,\ 12.5}_{\text{6 observations}},$$
$$\underbrace{17.5,\ 17.5}_{\text{2 observations}}).$$

The length is $4 + 8 + 6 + 2 = 20$. This sorted representative list has exactly the midpoint table's frequencies; it does not recover the original times or recording order.

Notation. Write $\bar{y}$, s_y^2, and s_y for the representative list's mean, sample variance, and sample standard deviation. Write $\bar{x}$, s_x^2, and s_x for the unknown statistics of the original observations. Thus $\bar{x} \approx \bar{y}$, $s_x^2 \approx s_y^2$, and $s_x \approx s_y$. The index i labels four classes; j labels twenty representative observations.

5. Sample mean

First state the list and equivalent weighted formulas:

$$\bar{y} = \frac{\sum_{j=1}^{n} y_j}{n} = \frac{\sum_{i=1}^{r} f_i m_i}{n}.$$

The product $f_i m_i$ adds midpoint m_i a total of f_i times.

m_i (s)	f_i	$f_i m_i$ (s)
2.5	4	$4(2.5) = 10$
7.5	8	$8(7.5) = 60$
12.5	6	$6(12.5) = 75$
17.5	2	$2(17.5) = 35$
Total	20	180

Substitute the midpoints and counts; divide by $n = 20$, not by four classes:

$$\bar{y} = \frac{4(2.5) + 8(7.5) + 6(12.5) + 2(17.5)}{20} = \frac{10 + 60 + 75 + 35}{20} = 9 \text{ s}.$$

Summing the expanded list also gives 180 seconds, so both representations give 9 seconds.

6. Sample variance

Use the reference convention: the sample denominator is $n - 1$.

$$s_y^2 = \frac{\sum_{j=1}^{n} (y_j - \bar{y})^2}{n-1} = \frac{\sum_{i=1}^{r} f_i (m_i - \bar{y})^2}{n-1}.$$

Subtract the unrounded mean 9 from each midpoint, square the difference, then multiply by the frequency to include every observation.

m_i	f_i	$m_i - 9$	$(m_i - 9)^2$	$f_i(m_i - 9)^2$
2.5	4	-6.5	42.25	$4(42.25) = 169$
7.5	8	-1.5	2.25	$8(2.25) = 18$
12.5	6	3.5	12.25	$6(12.25) = 73.5$
17.5	2	8.5	72.25	$2(72.25) = 144.5$

$$\sum f_i (m_i - 9)^2 = 169 + 18 + 73.5 + 144.5 = 405 \text{ s}^2, \qquad s_y^2 = \frac{405}{20 - 1} = \frac{405}{19} \text{ s}^2.$$

The list repeats these same squared deviations with the same frequencies, giving 405 as well. The denominator is 19, not the number of classes minus one.

7. Sample standard deviation

$$s_y = \sqrt{s_y^2}, \qquad s_y = \sqrt{\frac{405}{19}} \text{ s} \approx 4.617 \text{ s}.$$

Midpoint estimates for the original sample: $\bar{x} \approx 9.000$ s; $s_x^2 \approx 21.316$ s^2; $s_x \approx 4.617$ s. Mean and standard deviation have time units; variance has squared time units.

8. Absolute and percentage errors for C7

For estimate $\widehat{\theta}$ and nonzero theoretical benchmark θ, first use

$$E_\theta = |\widehat{\theta} - \theta|, \qquad R_\theta = \frac{E_\theta}{|\theta|} \times 100\%.$$

Absolute error is the size of the difference, in the statistic's units. Percentage error compares that difference with the theoretical value, not the sample estimate. All benchmarks here are nonzero. First find the theoretical standard deviation:

$$\sigma = \sqrt{\sigma^2}, \qquad \sigma = \sqrt{25} = 5 \text{ s}.$$

Mean: substitute $\bar{y} = 9$ for the estimate and $\mu = 10$ for the benchmark.

$$E_\mu = |9 - 10| = |-1| = 1 \text{ s}, \qquad R_\mu = \frac{1}{10} \times 100\% = 10.00\%.$$

Variance: substitute the exact fraction $405/19$ and $\sigma^2 = 25$.

$$E_{\sigma^2} = \left|\frac{405}{19} - 25\right| = \left|\frac{405 - 475}{19}\right| = \frac{70}{19} \text{ s}^2 \approx 3.684 \text{ s}^2,$$
$$R_{\sigma^2} = \frac{70/19}{25} \times 100\% = \frac{280}{19}\% \approx 14.74\%.$$

Standard deviation: substitute $\sqrt{405/19}$ and $\sigma = 5$. Because $405/19 < 25$, its square root is less than 5; hence

$$E_\sigma = \left|\sqrt{\frac{405}{19}} - 5\right| = \left(5 - \sqrt{\frac{405}{19}}\right) \text{ s} \approx 0.383 \text{ s},$$
$$R_\sigma = \frac{5 - \sqrt{405/19}}{5} \times 100\% \approx 7.66\%.$$

Keep the exact fraction and square root in error calculations. Round only the final statistics and absolute errors to three decimal places and percentage errors to two decimal places.

Statistic	Midpoint estimate	Theoretical value	Absolute error	Percentage error
Mean (s)	9.000	10.000	1.000	10.00%
Variance (s^2)	21.316	25.000	3.684	14.74%
Standard deviation (s)	4.617	5.000	0.383	7.66%

All three midpoint statistics are below their model benchmarks. These errors compare midpoint estimates with the model. They do not isolate the unknown error caused by grouping: that requires the original measurements. Sampling variation can also cause a sample to differ from its theoretical model.

9. Interval probability from the same dataset

The event $A = \{5 \le X < 15\}$ contains two complete classes, $[5, 10)$ and $[10, 15)$. Let k count files in the event.

$$k = \sum_{i:\,[L_i,U_i)\subseteq A} f_i, \qquad \widehat{P}(A) = \frac{k}{n}.$$

Substitute the heights of the included bars and the total count:

$$k = 8 + 6 = 14, \qquad n = 4 + 8 + 6 + 2 = 20, \qquad \widehat{P}(5 \le X < 15) = \frac{14}{20} = 0.70.$$

The interval table gives the same count. The list contains eight copies of 7.5 and six copies of 12.5 in the event: again 14 out of 20. Midpoint replacement changes no memberships here because the event contains whole classes. Thus 0.70 is the exact relative frequency of the original sample, while it is an empirical estimate of the process probability.

Compare with the theoretical probability $p = 0.80$. First state

$$E_p = |\widehat{P}(A) - p|, \qquad R_p = \frac{E_p}{|p|} \times 100\%.$$

Then substitute the sample proportion and the supplied benchmark:

$$E_p = |0.70 - 0.80| = 0.10, \qquad R_p = \frac{0.10}{0.80} \times 100\% = 12.50\%.$$

C6 result: empirical probability 0.70 (70%); absolute error 0.10, or 10 percentage points; relative percentage error 12.50%. Percentage points and percentage error are different comparisons.

10. Check what each representation preserves

Representation	Information and checks
Histogram and interval table	Same boundaries and frequencies; $n = 20$. Individual times are unknown.
Midpoint frequency table	Four representative values with frequencies $4, 8, 6, 2$. Count stays 20; original times are approximated.
Expanded list	Exactly the same representative data as the midpoint table: 20 entries, sum 180 s, squared-deviation sum 405 s^2.
Statistical calculations	List and weighted formulas agree: $\bar{y} = 9$, $s_y^2 = 405/19$, $s_y = \sqrt{405/19}$.
Interval probability	14/20 is exact for this sample because the event uses whole classes. A boundary cutting through a class would require more information or an extra approximation.

Interpretation. The midpoint data average 9 seconds per download with sample standard deviation about 4.617 seconds. These estimate the original sample's typical time and spread; model benchmarks describe the process. Changing representation does not create additional observations.